MIL-A-14819(MU)
21 October 1968

Superseding
FA(FC)-PD-73
9 May 1961

MILITARY SPECIFICATION
AMPLIFIER, AUDIO FREQUENCY: 10512508

1. SCOPE

1.1 This specification covers an Audio Amplifier used in Chronograph, M36. The Audio Amplifier is used to amplify and filter the detected doppler-frequency signal.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction, M18, Manufacturing Process and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWING

U. S. Army, Frankford Arsenal

F10512508	Amplifier, Audio Frequency: 10512508

PACKAGING DATA SHEET

10512508	Packaging of Amplifier, Audio Frequency

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The Audio Amplifier shall be manufactured in accordance with Drawing F10512508 and drawings pertaining thereto.

3.2 General Specification.- The Audio Amplifier shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of prededence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship
e. Soldering electrical connections

3.2.2 MIL-C-46357

a. Wiring
b. Threaded fasteners
c. Marking
d. Finishes
e. Etched circuit assemblies

3.3 Environmental

3.3.1 Shock.- The Audio Amplifier shall be capable of continuous operation while being subjected to three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

a. Bottom down
b. Front down
c. Right side down
d. Back down
e. Left side down
f. Top down

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Subsequent to the above shock, the Audio Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.3.2 Vibration.- The Audio Amplifier shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz. The entire frequency range, from 10 to 55 hertz and return to 10 hertz, shall be uniformly traversed in one (1) minute ± 15 seconds. This motion shall be applied for a period of thirty (30) minutes ± one (1) minute in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 1.5 hours. During the above vibration, the Audio Amplifier, with power applied, shall comply with 3.5.5, except, only the mid-frequencies of Column D Table I shall be monitored for a continuous uninterrupted undistorted signal at the levels required in Column E Table I. Subsequent to the above vibration, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.3 Humidity.- The Audio Amplifier shall be capable of continuous operation during four (4) hours ± ten (10) minutes of continuous exposure to a relative humidity of ninety (90) to ninety-five (95) percent at a temperature of plus 100° ± 5°F. During and subsequent to the above humidity, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.4 Sand and Dust.- The Audio Amplifier shall be capable of withstanding three (3) hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm and blowing dust of particle size 0.0001 to 0.01mm with a density between 0.1 and 0.5 gram per cubic foot of dust at a temperature of plus 75° ± 5°F, a relative humidity between fifteen (15) to thirty (30 percent, and a wind velocity of fifteen (15) knots. Subsequent to the above sand and dust exposure, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5 Altitude

3.3.5.1 Operating.- The Audio Amplifier shall be capable of continuous operation during four (4) hours of continuous exposure to an altitude of 10,000 feet at a temperature of plus 75°F. During and subsequent to the above exposure, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.5.2 Non-Operating.- The Audio Amplifier shall be capable of withstanding four (4) hours of continuous exposure to an altitude of 50,000 feet at a temperature of minus 65°F. Subsequent to the above exposure, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

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3.3.6 Fungus.- The Audio Amplifier shall be capable of withstanding twenty-eight (28) days exposure to a composite spore suspension consisting of the following cultures:

a. Chaetominum globosum
b. Aspergillus flavus
c. Aspergillus versicalor
d. Penicillum luteum
e. Rhizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of 80°F ± 4°F with an RH of 87 ± 2%. Subsequent to the above exposure, the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.7 Storage Temperatures.- The Audio Amplifier shall be capable of withstanding temperature stabilization at plus 160°F and minus 80°F for a period of eight (8) hours at each temperature extreme. Subsequent to each of the above temperature extremes (+160°F and -80°F) after temperature stabilization at +75° ± 15°, the Audio Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.3.8 Operating Temperatures.- The Audio Amplifier shall be capable of continuous operation for eight (8) hours at +140°F. The Audio Amplifier shall also be capable of continuous operation for eight (8) hours at minus 50°F. During and subsequent to the above temperature extremes (+140°F and -50°F), the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of 3.5 at least once at the beginning, middle, and ending of each eight (8) hour temperature extreme. There shall be a minimum of five (5) to six (6) minutes warm-up time prior to performing the first test of 3.5 at -50°F.

3.4 ELECTRICAL

3.4.1 Continuity.- The Audio Amplifier shall demonstrate continuity in accordance with the electrical schematic, Drawing F10525617.

3.5 PERFORMANCE

The performance requirements specified herein shall be met on every Audio Amplifier at Standard Ambient conditions prior to government acceptance.

3.5.1 Power Supply and Loads.- All requirements of 3.5 shall be met over the following power supply ranges and with the following loads except where a paragraph specifically indicates an exception:

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(a) +21.5 to +22.5 VDC - Pin 8
(b) 2K Ohms - Pin 4 to Pin 6
(c) Ground - Pin 6

3.5.2 Noise Output.- With the input grounded, the output shall not exceed 2 millivolts RMS with the Velocity Range switch set for those ranges given in Table I Column A.

3.5.3 AGC Output.

3.5.3.1 No Signal Input.- With no voltage input, the AGC output voltage shall be +8 ± 0.5 VDC.

3.5.3.2 Signal Input.- With a 20 millivolt RMS sine wave input at a frequency of 33KHz, the AGC output voltage shall have a minimum negative change of 0.3 VDC.

3.5.4 Signal Output Level Adjustment.- With the Velocity Range switch in the 70-325 position, inject a 1.5 millivolt RMS signal of 12KHz. Adjust and lock R17 to obtain an undistorted sine wave output of 200 millivolts peak-to-peak at 12KHz. This adjustment shall be performed with the 22.0 VDC (nominal value) set at 21.5 VDC.

3.5.5 Frequency Bandpass Response.- As delineated in Table I, with the Velocity Range switch in the proper position (Column A) and the corresponding undistorted sine wave Input (Column B) over the corresponding Frequency Bandpass Range (Column C) the corresponding undistorted sine wave Output (Column E) shall be greater than the minimum value indicated.

TABLE I

A Velocity Range Switch Position	B Sine Input (MV RMS)	C Freq Bandpass Range (KHz)	D Midband Freq (KHz)	E Sine Output Minimum (MV P-P)
70 - 325	1.5	12 - 23	17.5	200
300 - 400	1.5	21 - 29	25	200
375 - 525	1	26.3 - 37	31.6	250
500 - 700	1	34 - 50	42	250
650 - 800	1	45 - 57	51	250
750 - 1000	1	52 - 71	61.5	250
925 - 1150 CAL	1	63 - 81	72	250
1100 - 1400	1	76 - 98	87	250
1400 - 1590	1	100 - 113	106.5	250
1525 - 1800	1	106 - 126	116	250

3.5.6 Frequency Bandreject Response.- As delineated in Table II, with the Velocity Range switch in the proper position (Column A) and the corresponding undistorted sine wave Input (Column B) over the corresponding Frequency Bandreject Ranges (Column C) the corresponding Total Output (Column D) shall be no greater than the maximum value indicated.

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TABLE II

A Velocity Range Switch Position	B Sine Input (MV RMS)	C Freq. Bandreject Ranges (KHz)	D Total Output Maximum (MV P-P)
70 - 325	30	0-3 35-150	30
300 - 400	30	0-12 42-150	30
375 - 525	30	0-18 49-150	30
500 - 700	30	0-23 63-150	30
650 - 800	30	0-33 78-150	30
750 - 1000	30	0-40 90-150	30
925 - 1150 CAL	30	0-50 100-150	30
1100 - 1400	30	0-60 110-150	30
1400 - 1590	30	0-100 140-150	30
1525 - 1800	30	0-100 140-150	30

3.5.7 Distortion.- As delineated in Table III, with the Velocity Range switch in the proper position (Column A) and an undistorted sine wave Input varying over the ranges given (Column B) at the Midband Freq (Column C) the output shall be an undistorted sine wave at the Midband Freq (Column C).

TABLE III

A Velocity Range Switch Position	B Sine Input Range (MV RMS)	C Midband Freq (KHz)
70 - 325	1.5 - 100	17.5
300 - 400	1.5 - 100	25.0
375 - 525	1 - 100	31.6
500 - 700	1 - 100	42.0
650 - 800	1 - 100	51.0
750 - 1000	1 - 100	61.5
925 - 1150 CAL	1 - 100	72.0
1100 - 1400	1 - 100	87.0
1400 - 1590	1 - 100	106.5
1525 - 1800	1 - 100	116.0

3.6 Reliability Run-In.- In order to maximize component reliability, prior to acceptance by the government, each Audio Amplifier shall be subject to the following run-in conditions:

3.6.1 Temperature Run-In.- The Audio Amplifier, without power applied, shall be temperature stabilized at plus 125°F and minus 25°F. At each temperature extreme, after temperature stabilization, power shall be applied and each unit shall be tested to all the performance requirements of 3.5 for a period of three (3) hours with a minimum of one (1) complete

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test to the requirements of 3.5 occurring at the beginning, middle, and end of the three (3) hour period. There shall be a five (5) to six (6) minute warm-up time prior to the first performance test at -50°F. Subsequent to the above tests, the Audio Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.6.2 Vibration Run-In.- The Audio Amplifier shall be capable of continuous operation while being vibrated in a simple harmonic motion having an amplitude of 0.03 (0.06 inch maximum total excursion) with the frequency being varied uniformly between the limits of ten (10) and fifty-five (55) hertz and return to ten (10) hertz and shall be uniformly traversed in one (1) minute. This motion shall be applied for a period of ten (10) minutes in each of three (3) mutually perpendicular directions (x, y, z axis) for a total of 0.5 hours. During the above vibration, the Audio Amplifier with power applied, shall meet the requirements of 3.5.5; except, only the Midband Frequencies of Column D, Table I shall be monitored for a continuous uninterrupted undistorted Sine Output signal at the levels required in Column E, Table I. Subsequent to the above vibration, the Audio Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

3.6.3 Shock Run-In.- The Audio Amplifier shall be capable of three (3) shocks of 50 gravity units (g's) of eleven (11) millisecond duration in each of the following directions:

a. Bottom Down
b. Front Down
c. Right Side Down
d. Back Down
e. Left Side Down
f. Top Down

During the above 18 shocks, the Audio Amplifier with power applied, shall meet the requirements of 3.5.5; except, only the Midband Frequencies of Column D, Table I shall be monitored for a continuous uninterrupted undistorted Sine Output signal at the levels required in Column E, Table I. The velocity range switch shall be in the 70-325 position on the first shock and progressively increase to the 1525-1800 position on the tenth shock. The eleventh shock shall be progressively repeated as above.

Subsequent to the above shock, the Audio Amplifier shall not have been damaged or adversely affected, and shall meet the requirements of 3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (Initial Production) Approval.- The requirement for first article approval and responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) audio amplifiers manufactured in the same manner using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Examination and Tests

4.3.1.1 Components and Subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2 Acceptance and Rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling.- At the beginning of production and subsequent to first article approval (see 4.2), three units shall be selected by a Government representative and two from each fifty (50) produced. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

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CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
401. Fabrication	3.1	Applic. Dwgs & Specs - Visual
402. General Specifications	3.2	MIL-F-13926 & MIL-C-46357
403. Shock	3.3.1	4.6.1.1
404. Vibration	3.3.2	4.6.1.2
405. Humidity	3.3.3	4.6.1.3
406. Sand and Dust	3.3.4	4.6.1.4
407. Altitude-operating	3.3.5.1	4.6.1.5
408. Altitude-Non-operating	3.3.5.2	4.6.1.5.1
409. Storage Temperatures	3.3.7	4.6.1.7.1
410. Operating Temperatures	3.3.8	4.6.1.7.2

4.4.1 Failure of Sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor, and resubmitted samples have been approved (see 4.3.2).

4.5 Inspection Equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government Furnished Inspection Equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment.-

4.5.2.1 Contractor Design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification, and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods and Procedures.

4.6.1 Environmental.- Test equipment and test facilities utilized in the following environmental tests shall be in accordance with "Test Facilities" requirements of MIL-F-13926 and conditions specified in 3.3.

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4.6.1.1 Shock.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.6.3. Subsequent to the eighteen (18) shocks, the Audio Amplifier shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.2 Vibration.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the conditions specified in 3.3.2. During the above vibration, the Audio Amplifier shall meet the performance tests as specified in 4.6.2.4.1, except that only the mid-frequencies of Column D Table I shall be monitored for a continuous uninterrupted undistorted signal at the levels required in Column E Table I. Subsequent to the 1.5 hour vibration, the Audio Amplifier shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.3 Humidity.- The humidity test shall be performed as specified in MIL-STD-810, Method 507, Procedure I for four (4) hours at the conditions specified in 3.3.3. Subsequent to this test, the Audio Amplifier shall be examined for damage then tested as specified in 4.6.2.

4.6.1.4 Sand and Dust.- The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.3.4. Subsequent to this test, the Audio Amplifier shall be examined for damage then tested as specified in 4.6.2.

4.6.1.5 Altitude-Operating.- The altitude test shall be performed as specified in MIL-F-13926, Procedure I at an altitude of 10,000 feet and as specified in 3.3.5.1. The Audio Amplifier shall be exposed to the above condition for a time period required to allow all entrapped air in the chamber to escape as evidenced by chamber pressure stabilization. The barometric pressure shall then be maintained to simulate 10,000 feet (20.58 inches of mercury) altitude condition. While the Audio Amplifier is maintained at this pressure for four (4) hours, the Audio Amplifier shall be subjected to the tests specified in 4.6.2.

4.6.1.5.1 Altitude Non-Operating.- The Audio Amplifier shall be subjected to barometric pressure specified in MIL-F-13926, Procedure II for four (4) hours as specified in 3.3.5.2. Subsequent to this test, the Audio Amplifier shall be examined for damage, then tested as specified in 4.6.2.

4.6.1.6 Fungus.- One (1) Audio Amplifier shall be subjected to the fungus test specified in MIL-F-13926. At the end of the 28 day incubation period, the Audio Amplifier shall be tested as specified in 4.6.2 for compliance with 3.3.6. Subsequent acceptance of Audio Amplifiers shall be by certification.

4.6.1.7 Temperature Cycling

4.6.1.7.1 Storage Temperatures.- This test shall be performed as specified in MIL-F-13926, Procedure II, and thermally stabilized at +160°F for eight (8) hours and at -80°F for eight (8) hours. The Audio Amplifier shall

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be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.1.7.2 Operating Temperatures.- The Audio Amplifier shall be placed in a chamber and the temperature gradually raised until thermally stabilized at +140°F for eight (8) hours. The Audio Amplifier shall be tested as specified in 4.6.2 during the first hour, at the four (4) hour period, and within the last hour of the eight (8) hour period. The temperature shall then be reduced to -50°F. When thermally stabilized at this temperature, the Audio Amplifier shall be tested as specified in 4.6.2 at the three (3) time periods specified above. A minimum of five (5) to six (6) minutes warm-up period shall occur prior to performance of the first test. The Audio Amplifier shall be returned to room ambient temperature +75 ± 15°F, examined for damage then tested as specified in 4.6.2.

4.6.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily accomplished on every Audio Amplifier at Standard Ambient Conditions. Standard Ambient Conditions for the purposes of these tests shall be defined as:

a. Temperature between 60°F and 90°F

b. Barometric pressure between 28 and 32 inches of mercury

c. Relative humidity (RH) up to 90%

4.6.2.1 Power Supply and Load Conditions.- All requirements of paragraph 4.6.2 shall be met under the following power supply and load conditions except where an exception is specifically delineated.

Connection	Condition I	Nominal	Condition II
P1-8	21.5 VDC	22.0 VDC	22.5 VDC
P1-4	1.9K Load	2.0K Load	2.1K Load
P1-6	Ground	Ground	Ground

4.6.2.2 Noise Output.- Ground P1-1 and monitor with an oscilloscope the output across P1-4 to ground (P1-6) with the Velocity Range switch in the 70-325 position. The total output voltage shall not exceed 5.6 millivolts peak-to-peak. Repeat the above test for the nine (9) remaining Velocity Range switch positions as given in Table I, Column A.

4.6.2.3 AGC Output.- The AGC Output shall be measured between P1-3 and P1-6.

4.6.2.3.1 No Signal Input.- With P1-1 grounded, the AGC Output shall be +8.0 ± 0.5 VDC.

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4.6.2.3.2 Signal Input.- With a 20 millivolt RMS signal at 33KHz applied to P1-1, the AGC Output voltage shall have a minimum negative change of 0.3 VDC.

4.6.2.4 Signal Output Level Adjustment.- With the Velocity Range switch in the 70-325 position, inject an undistorted sine wave input at P1-1 of 1.5 millivolts RMS at a frequency of 12 KHz, adjust and lock R17 to obtain an undistorted sine wave output at P1-4 of 200 millivolts peak-to-peak. This adjustment shall be accomplished with the 22.0VDC (norminal value) set at 21.5VDC.

4.6.2.4.1 Frequency Bandpass Response.- With the Velocity Range switch in the 70-325 position, inject a 1.5 millivolt Sine Input signal at P1-1 and sweep the Frequency Bandpass Range of 12 to 23KHz. The Sine Output voltage at P1-4 shall be an undistorted sine wave having a frequency corresponding to the input frequency and an amplitude equal to or greater than 200 millivolts peak-to-peak. Repeat the above procedure using the remaining Velocity Range switch positions, their associated Sine Input values, their associated Frequency Bandpass ranges, and their associated Sine Output minimums as given in Table I.

4.6.2.4.2 Frequency Bandreject Response.- With the Velocity Range switch in the 70-325 position, inject a 30 millivolt RMS Sine Input signal at P1-1 and sweep the Frequency Bandreject Ranges of 0 to 3 and 35 to 150KHz. The Total Output voltage at P1-4 shall be less than 30 millivolts peak-to-peak. Repeat the above procedure using the remaining Velocity Range switch positions, their associated Sine Input values, their associated Frequency Bandreject Ranges, and their associated Total Output maximums as given in Table II.

4.6.2.4.3 Distortion.- With the Velocity Range switch in the 70-325 position, inject 1.5 millivolt RMS Sine Input signal at the Midband frequency of 17.5KHz at P1-1. The Sine Output at P1-4 shall be an undistorted sine wave having a frequency and a amplitude greater than 200MV peak-to-peak. Increase the Sine Input from 1.5MV RMS to 100MV RMS, the output shall be an undistorted sine wave at the same frequency as the input frequency over the entire voltage input range. Repeat the above procedure using the remaining Velocity Range switch positions, their associated Sine Input range values, their Midband Frequency to obtain an undistorted sine wave output at P1-4 at their corresponding midband frequency as given in TABLE III.

4.6.3 Electrical

4.6.3.1 Continuity.- Continuity shall be tested by standard measuring equipment for compliance with 3.4.1.

4.6.4 Reliability Run-In.- The following Run-In shall be performed on a 100% basis.

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4.6.4.1 Temperature Run-In.- The Audio Amplifier shall be placed in a chamber and the temperature gradually raised until thermally stabilized at plus 125°F. After temperature stabilization, power shall be applied and maintained for three (3) hours. Five (5) minutes after power is applied, a complete test to the requirements of paragraph 4.6.2 shall commence. A second test to the requirements of paragraph 4.6.2 shall be accomplished during the second hour and a third test to the requirements of paragraph 4.6.2 shall be accomplished during the third hour of the three (3) hour test. Repeat the above tests at a temperature of minus 25°F. Subsequent to the above temperature extremes (plus 125°F and minus 25°F) the Audio Amplifier shall not have been damaged or adversely affected and shall meet the requirements of paragraph 4.6.2.

4.6.4.2 Vibration Run-In.- The vibration test shall be performed as specified in MIL-STD-810, Method 514 at the vibration conditions specified in 3.6.2. During the above vibration test, the Audio Amplifier shall meet the performance tests of 4.6.2; except 4.6.2.4 shall be accomplished prior to the start of the vibration test. When performing 4.6.2.4.1 and 4.6.2.4.2, the mid-frequencies of the Freq. Bandpass Range (Table I, Column C) and Freq. Bandreject Ranges (Table II, Column C) shall be used in lieu of sweeping the full frequency ranges. Subsequent to the vibration test, the Audio Amplifier shall be examined for damage and shall not be damaged or adversely affected and shall meet the performance requirements of 4.6.2.

4.6.4.3 Shock Run-In.- The shock test shall be performed as specified in MIL-STD-810, Method 516, Procedure IV at the conditions specified in 3.6.3. Subsequent to the eighteen (18) shocks, the Audio Amplifier shall be examined for damage and shall not be damaged or adversely affected and shall meet the requirements of 4.6.2.

4.6.4.4 Failures.- Failures occurring during Reliability Run-In shall not be cause for rejection, but shall be corrected as outlined in paragraph 3.6.4.

4.6.5 Packaging, Packing, and Marking.- Quality Assurance Provisions for packaging, packing and marking shall be specified in MIL-P-14232 and Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10512508. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use.- The Audio Amplifier is used to amplify the detected doppler-frequency signal to a level suitable for operation of the indicator.

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6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock number
d. Applicable packaging data sheet (see 5.1)

Custodian:

Army MU

Preparing Activity:

Army MU

Project No. 1290-A198

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Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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